A superquadratic separation between sensitivity and block sensitivity

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Abstract

We disprove the quadratic strengthening of the Sensitivity Conjecture by constructing nonconstant total Boolean functions whose block sensitivity grows faster than any constant multiple of sensitivity squared. In fact, for some fixed $\alpha > 2$, the examples have unbounded block sensitivity and satisfy $\mathrm{bs}(f) \geq s(f)^\alpha$.

1 The result

1.1 Definitions and main theorem

For a positive integer n write $[n] = \{1, \ldots, n\}$. For $x \in \{0,1\}^n$ and $B \subseteq [n]$, let x^B be x with exactly the coordinates in B flipped, and write $x^e = x^{\{e\}}$. Let $f : \{0,1\}^n \to \{0,1\}$ be a total Boolean function, that is, one defined on all of $\{0,1\}^n$.

- *Sensitivity:* $s(f,x) = \left|\{e \in [n] : f(x^e) \neq f(x)\}\right|$ and $s(f) = \max_x s(f,x)$.
- *Block sensitivity:* a nonempty $B \subseteq [n]$ is a *sensitive block at* x if $f(x^B) \neq f(x)$. Then $\mathrm{bs}(f,x)$ is the largest number of pairwise disjoint sensitive blocks at x, and $\mathrm{bs}(f) = \max_x \mathrm{bs}(f,x)$.

Theorem 1.1. *For every real $C > 0$, there are a positive integer n and a nonconstant total Boolean function $f : \{0,1\}^n \to \{0,1\}$ such that*

$$\mathrm{bs}(f) > C\, s(f)^2.$$

More precisely, for every integer $d \geq 1$ the construction below gives a nonconstant total Boolean function f with

$$\frac{\mathrm{bs}(f)}{s(f)^2} \geq \frac{2^d}{4(d+2)^2}.$$

Corollary 4.3 upgrades this to $\mathrm{bs}(F_m, 0) \geq s(F_m)^\alpha$ for a fixed $\alpha > 2$ and a sequence F_m with unbounded block sensitivity.

Background. Whether block sensitivity is bounded by a polynomial in sensitivity was the Sensitivity Conjecture. It goes back to Nisan's work on parallel computation, which introduced block sensitivity and recorded Rubinstein's quadratic separation [Nis89, Section 2], [Rub95]. Nisan and Szegedy suggested $\mathrm{bs}(f) \leq s(f)^2$ [NS94, Section 4]. Virza obtained $\mathrm{bs}(f) = \frac{1}{2}s(f)^2 + \frac{1}{2}s(f)$ [Vir11], and Ambainis and Sun $\mathrm{bs}(f) = \frac{2}{3}s(f)^2 - \frac{1}{3}s(f)$ [AS11, Theorem 1], which left a universal quadratic bound open. Huang's degree bound with Tal's refinement [Tal13] gives $\mathrm{bs}(f) \leq s(f)^4$ [Hua19, Theorem 1.5], and Wellens proved $\mathrm{bs}(f) \leq \sqrt{2/3}\, s(f)^4 + 1$ [Wel22, Corollary 4.2]. Theorem 1.1 rules

out the quadratic strengthening. The construction adapts the tournament-indexed rows and selected blocking coordinates of Meiburg's spectral-sensitivity construction [Mei26, Sections 3.1 and 5.2–5.5],[1] with a different forbidden-label condition. Spectral sensitivity $\lambda(f)$ [ABDK+21] satisfies $\lambda(f) \le s(f)$, so a lower bound on $\mathrm{bs}(f)/\lambda(f)^2$ alone would not give Theorem 1.1; no spectral input is used below. The final OR step follows Ambainis and Sun [AS11, Section 4, Lemma 1]. The amplification of a seed with $\mathrm{bs} > s^2$ by iteration was noted by Ambainis and Prūsis [AP14, Section 1] and is an instance of the composition method of Tal [Tal13]. None of these is used as a black box: every step is proved here.

1.2 Proof at a glance

Fix a depth d and a large parameter M; put $k = 2M^2 + 1$ and $r = \lceil \sqrt{M} \rceil$.

Construction (Section 2). For each level $\ell = 0, \dots, d$ we define predicates $P_{\ell,1} \ge P_{\ell,2} \ge \cdots$ (pointwise; "nested"). At level 0 they are indicators of unions of Hamming balls of radii $d, d-1, \dots, 0$ around M far-apart centres. A level-ℓ input is a $k \times r$ array of level-$(\ell-1)$ inputs ("children"). The k rows are the vertices of a tournament with all outdegrees M^2, and each edge $i \to j$ carries a label $a(i,j) \in [r]$ pointing at one child of row j. The predicate $P_{\ell,q}$ accepts when some *clause* i holds: the strongest child predicate accepts every child of row i (*targets*), and a weaker child predicate, depending on q, rejects the labelled child in each row j with $i \to j$ (*gates*).

Many blocks (Lemma 2.3). At the all-zero input, switching on one row satisfies a clause, so the number of disjoint sensitive blocks is multiplied by $k > 2M^2$ per level.

Few sensitive bits (Proposition 3.1). Let $S_{\ell,b}$ bound the number of single-bit flips taking a level-ℓ predicate from b to $1-b$, and let $J_{\ell,b}$ bound the flips that move *two distinct* predicates from b to $1-b$ at once. Then

$$S_{\ell,1} \le M^2 S_{\ell-1,0} + r S_{\ell-1,1}, \qquad S_{\ell,0} \le 16 S_{\ell-1,0} + S_{\ell-1,1} + 3 J_{\ell-1,1}.$$

For the first, fix one true clause; a flip must break one of its M^2 gates or r targets. The second is the heart of the proof. A flip from 0 to 1 must complete a clause that currently fails exactly one condition. The labels (Lemma 2.1) allow at most 15 clauses missing one target. The tournament allows at most 3 clauses missing one gate, and all but one of these can be completed only by a flip changing two child predicates simultaneously. The J's obey similar recurrences and vanish at level 0 (Lemma 2.2), because one bit cannot cross two radii.

Bookkeeping (Section 4). For $M \ge 9^{d+1}$ the recurrences give $S_{\ell,b} \le 2(d+2)M^{\ell+b}$ (Lemma 4.1): sensitivity grows by a factor about M per level, blocks by more than $2M^2$. The OR f of M copies of $P_{d,1}$ has $s(f) \le 2(d+2)M^{d+1}$ and $\mathrm{bs}(f,0) \ge M^2 k^d$, which is Theorem 1.1. Composing the $d = 9$ instance with itself (Lemma 4.2) gives Corollary 4.3.

Dependencies. Lemma 2.1 ⇒ Proposition 3.1; Lemma 2.2 and Proposition 3.1 ⇒ Lemma 4.1; Lemmas 2.3 and 4.1 ⇒ Theorem 1.1; the $d = 9$ case and Lemma 4.2 ⇒ Corollary 4.3.

Notation changes from the source. A coordinate is called e (the source's a clashed with the labels $a(i,j)$). In Lemma 4.2 the dimensions are n_1, n_2, the block counts p_1, p_2 and the blocks Γ_i, Δ_j, to avoid clashes with q, b, C and E_j.

[1] In the attribution accompanying that construction, Meiburg credits GPT-5.6-Sol with the gated-tournament ideas and describes his own contributions as exposition, size and parameter optimization, and checking the faithfulness of the Lean formalization.

2 The labelled tournament and nested predicates

2.1 Parameters and labels

Fix integers $d \geq 1$ and $M \geq 3$, and put

$$L = d + 2, \qquad k = 2M^2 + 1, \qquad r = \lceil \sqrt{M} \rceil, \qquad t = 16.$$

(Section 4 takes $M \geq 9^{d+1}$.) A *tournament* on $[k]$ contains, for each pair of distinct $i, j \in [k]$, exactly one of the directed edges $i \to j$, $j \to i$. Fix one in which every vertex has outdegree M^2. One exists: identify $[k]$ with $\mathbb{Z}/k$ and let $i \to j$ when $j - i \in \{1, \dots, M^2\}$ modulo k; since $k = 2M^2 + 1$, exactly one of $j - i$ and $i - j$ lies in that range.

Lemma 2.1. *The edges can be labelled by numbers $a(i, j) \in [r]$, one for each edge $i \to j$, so that there is no set $I \subseteq [k]$ of size t admitting choices $m_j \in [r]$, $j \in I$, with*

$$a(i, j) = m_j \qquad \textit{for every edge } i \to j \textit{ within } I.$$

This is used only to bound $|\mathcal{T}|$ in Step 4 of Proposition 3.1.

Proof. Choose all labels independently and uniformly from $[r]$. Call a pair $(I, (m_j)_{j \in I})$ a *pattern*; it is *realised* if all the displayed equalities hold.

One pattern. There are $\binom{16}{2} = 120$ edges within I. Each edge $i \to j$ imposes one equation $a(i, j) = m_j$ on its own label, which holds with probability $1/r$. Labels of different edges are independent, so the pattern is realised with probability r^{-120}.

Union bound. There are at most k^{16} sets I and r^{16} choices of (m_j). Hence

$$\Pr[\text{some pattern is realised}] \leq k^{16} r^{16} r^{-120} = k^{16} r^{-104} \leq (3M^2)^{16} (\sqrt{M})^{-104} = 3^{16} M^{-20} \leq 3^{-4} < 1,$$

using $k \leq 3M^2$, $r \geq \sqrt{M}$, $M^{32-52} = M^{-20}$ and $M \geq 3$. So some labelling realises no pattern. □

Fix such a labelling from now on.

2.2 The predicates

For $0 \leq \ell \leq d$, set

$$H_\ell = d + 1 - \ell, \qquad N_\ell = LM(kr)^\ell.$$

We define $P_{\ell,q} : \{0,1\}^{N_\ell} \to \{0,1\}$ for $1 \leq q \leq H_\ell$ by recursion on ℓ.

Level 0. Partition $[LM]$ into blocks $E_1, \dots, E_M$ of size L, and let $z_j \in \{0,1\}^{LM}$ be the indicator of E_j (the *centres*). With dist the Hamming distance, set

$$D(x) = \min_{j \in [M]} \operatorname{dist}(x, z_j), \qquad P_{0,q}(x) = \mathbf{1}\{D(x) \leq d + 1 - q\} \quad (1 \leq q \leq d + 1).$$

The radii $d + 1 - q$ run through $d, d - 1, \dots, 0$; distinct centres are at distance $2L = 2d + 4$.

Level $\ell \geq 1$. Put $H = H_\ell$ and $Q = H + 1 = H_{\ell-1}$, so Q is the largest index at level $\ell - 1$. Split $x \in \{0,1\}^{N_\ell}$ into kr disjoint *children* $x_{j,c} \in \{0,1\}^{N_{\ell-1}}$, with row $j \in [k]$ and position $c \in [r]$; this matches $N_\ell = krN_{\ell-1}$. For $i \in [k]$ and $q \in [H]$, say that *clause i holds at index q* (at x) if

$$\begin{aligned} P_{\ell-1,Q}(x_{i,c}) &= 1 \quad \text{for all } c \in [r] \qquad (\textit{target conditions}), \\ P_{\ell-1,Q-q}(x_{j,a(i,j)}) &= 0 \quad \text{for all } j \text{ with } i \to j \quad (\textit{gate conditions}). \end{aligned} \tag{1}$$

Define $P_{\ell,q}(x) = 1$ if some clause holds at index q, and 0 otherwise. The indices are valid because $1 \leq Q - q \leq H < Q$. A clause has r targets and M^2 gates; the targets do not depend on q, and the child tested by each gate does not depend on q.

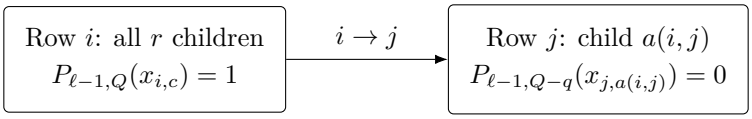


Figure 1: Clause i: one target per child of row i, one gate per out-neighbour j. Raising q strengthens the gates only.

Two facts are used throughout.

(N) Nesting. *For every level ℓ, input x and indices $p < p'$: if $P_{\ell,p'}(x) = 1$ then $P_{\ell,p}(x) = 1$; equivalently, if $P_{\ell,p}(x) = 0$ then $P_{\ell,p'}(x) = 0$. For $\ell \geq 1$, a clause holding at index p' also holds at index p.*

Proof by induction on ℓ. At level 0, $D(x) \leq d + 1 - p'$ implies $D(x) \leq d + 1 - p$. Let $\ell \geq 1$ and let clause i hold at index p'. Its targets are the same at index p. For each gate child y we have $P_{\ell-1,Q-p'}(y) = 0$ and $Q - p' < Q - p$, so (N) at level $\ell - 1$ gives $P_{\ell-1,Q-p}(y) = 0$. Hence clause i holds at index p, and $P_{\ell,p}(x) = 1$. □

Nesting concerns the predicate index, not the input bits: the smallest index is the easiest to satisfy.

(SC) One flip, one condition. *Fix $\ell \geq 1$, an index q and a clause i. Flipping one coordinate $e \in [N_\ell]$ changes the truth value of at most one condition of clause i at index q.*

Proof. The $r + M^2$ conditions test pairwise distinct children: the targets are the r children of row i; each gate tests a child in a row $j \neq i$, and distinct gates use distinct rows because there is only one edge between i and j. The coordinate e lies in exactly one child. □

The same flip may affect many clauses and several predicates of its child; the counts below are unions over clauses.

2.3 Sensitivity profiles and the initial bounds

For $b \in \{0,1\}$ and $0 \leq \ell \leq d$ define

$$S_{\ell,b} = \max_{\substack{1 \leq q \leq H_\ell,\ x \\ P_{\ell,q}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = 1 - b\}\big|,$$

$$J_{\ell,b} = \max_{\substack{1 \leq q < q' \leq H_\ell,\ x \\ P_{\ell,q}(x) = P_{\ell,q'}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1 - b\}\big|.$$

A maximum over no cases is zero; in particular $J_{d,0} = J_{d,1} = 0$, since $H_d = 1$. In J, both predicates are evaluated at the same input and changed by the same flip.

Lemma 2.2. *The initial sensitivity profiles satisfy*

$$S_{0,0} \le L, \qquad S_{0,1} \le LM, \qquad J_{0,0} = J_{0,1} = 0. \tag{2}$$

This is the base case of Lemma 4.1.

Proof. *(i)* $|D(x) - D(x^e)| \le 1$. For each j, the distances $\mathrm{dist}(x, z_j)$ and $\mathrm{dist}(x^e, z_j)$ differ by exactly 1, so the minima over j differ by at most 1.

(ii) $S_{0,0} \le L$. Fix q, put $R = d + 1 - q \in \{0, \dots, d\}$, and let $D(x) > R$. Suppose $D(x^e) \le R$, and pick a centre z_j with $\mathrm{dist}(x^e, z_j) \le R$. Then $\mathrm{dist}(x, z_j) \le R + 1 \le d + 1$. At most one centre lies within distance $d + 1$ of x, because two such centres would be at distance at most $2d + 2 < 2L$ by the triangle inequality. So z_j is the same centre z for every such e. Since $\mathrm{dist}(x, z) \ge D(x) \ge R + 1$, we get $\mathrm{dist}(x, z) = R + 1$, and the flip must decrease this distance: e is one of the $R + 1$ coordinates where x and z differ. Thus there are at most $R + 1 \le d + 1 \le L$ such e.

(iii) $S_{0,1} \le LM$, the number of coordinates.

(iv) $J_{0,b} = 0$. Let $q < q'$, with integer radii $R = d + 1 - q > R' = d + 1 - q'$. If a flip took both predicates from 0 to 1, then $D(x) > R \ge R' + 1$, so $D(x) \ge R' + 2$, while $D(x^e) \le R'$; this contradicts (i). If a flip took both from 1 to 0, then $D(x) \le R'$ and $D(x^e) \ge R + 1 \ge R' + 2$, again contradicting (i). □

2.4 Disjoint blocks at zero

Lemma 2.3. *At every level $0 \le \ell \le d$, all predicates are zero at the all-zero input. There are Mk^ℓ pairwise disjoint nonempty blocks of coordinates, each of size Lr^ℓ, such that flipping any one block at zero makes every predicate at that level equal to one.*

This gives the block-sensitivity lower bound in the proof of Theorem 1.1.

Proof. Induction on ℓ.

Level 0. Each centre has weight L, so $D(0) = L > d$ and every $P_{0,q}(0) = 0$. The M blocks E_j have size L, and $0^{E_j} = z_j$ has $D = 0$, so every $P_{0,q}(z_j) = 1$.

Level $\ell \ge 1$. Let $B_1, \dots, B_{Mk^{\ell-1}}$ be the blocks at level $\ell - 1$. At the all-zero input every child is all-zero, so $P_{\ell-1,Q} = 0$ on every child by induction; every clause fails a target, and every $P_{\ell,q}(0) = 0$.

For a row i and an index h, let $U_{i,h}$ be the union over $c \in [r]$ of the copy of B_h inside child (i, c). After flipping $U_{i,h}$ at zero, every child of row i equals 0^{B_h} and all other children are all-zero. By induction $P_{\ell-1,Q}(0^{B_h}) = 1$, so the targets of clause i hold. Each gate child of clause i lies in another row, is all-zero, and so has $P_{\ell-1,Q-q} = 0$ by induction; the gates hold for every q. Hence every $P_{\ell,q}$ equals 1 after the flip.

The sets $U_{i,h}$ are pairwise disjoint: for different rows they lie in different children, and for the same row they are disjoint inside each child because the B_h are disjoint. There are $k \cdot Mk^{\ell-1} = Mk^\ell$ of them, each of size $r \cdot Lr^{\ell-1} = Lr^\ell$. □

3 Sensitivity recurrences

Proposition 3.1. *For $1 \le \ell \le d$, write $S'_b = S_{\ell-1,b}$ and $J'_b = J_{\ell-1,b}$. Then*

$$\begin{aligned} S_{\ell,1} &\le M^2 S'_0 + r S'_1, & J_{\ell,1} &\le M^2 J'_0 + r S'_1, \\ S_{\ell,0} &\le t S'_0 + S'_1 + 3J'_1, & J_{\ell,0} &\le t S'_0 + 3J'_1. \end{aligned} \tag{3}$$

These four inequalities feed Lemma 4.1. They hold at every input, not only near the all-zero input.

Proof. Plan. On the one-side (Steps 1–2) we fix one true clause and count the flips that break it. On the zero-side (Steps 3–7) we show that only boundedly many clauses can be completed by a single flip, and count the flips completing each.

Step 0: conventions. Fix ℓ, $H = H_\ell$, $Q = H + 1$. A *flip* is a flip of one coordinate $e \in [N_\ell]$; it lies in one child y. By the definitions of S'_b and J'_b, applied to y:

(C1) for a fixed child y and index p, at most S'_b flips change $P_{\ell-1,p}(y)$ from b to $1-b$;

(C2) for a fixed child y and indices $p \neq p'$, at most J'_b flips change both $P_{\ell-1,p}(y)$ and $P_{\ell-1,p'}(y)$ from b to $1-b$.

Step 1: $S_{\ell,1} \leq M^2 S'_0 + r S'_1$. Fix q and x with $P_{\ell,q}(x) = 1$, and fix one clause i holding at index q at x. If $P_{\ell,q}(x^e) = 0$, then clause i fails at x^e, so the flip makes one of its conditions false. Either $P_{\ell-1,Q}$ changes from 1 to 0 on one of the r target children (at most S'_1 flips each, by (C1)), or $P_{\ell-1,Q-q}$ changes from 0 to 1 on one of the M^2 gate children (at most S'_0 flips each).

Step 2: $J_{\ell,1} \leq M^2 J'_0 + r S'_1$. Fix $q < q'$ and x with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 1$. Fix one clause i holding at index q' at x; by (N) it also holds at index q. Put

$$g = Q - q, \qquad g' = Q - q', \qquad 1 \leq g' < g < Q.$$

Let e be a flip after which both parent predicates are 0. Then clause i fails at index q at x^e.

Case A: a target of clause i fails at x^e. As in Step 1, at most $r S'_1$ flips.

Case B: all targets hold at x^e. Then some gate of clause i fails at index q: its child y has $P_{\ell-1,g} = 1$ after the flip, hence $P_{\ell-1,g'} = 1$ after the flip by (N). Before the flip, $P_{\ell-1,g'}(y) = 0$ because clause i held at index q', hence $P_{\ell-1,g}(y) = 0$ by (N). So the flip changes the two distinct predicates $P_{\ell-1,g'}$ and $P_{\ell-1,g}$ on y from 0 to 1. By (C2) there are at most J'_0 such flips for each of the M^2 gate children.

Step 3: candidate clauses at a zero. Fix q and x with $P_{\ell,q}(x) = 0$. Let $\mathcal{T}$ be the set of $i \in [k]$ whose clause, at index q at x, fails exactly one target and no gate, and $\mathcal{G}$ the set of i whose clause fails exactly one gate and no target. Both sets depend only on (x, q), not on any flip.

Claim. If clause i holds at index q at x^e, then $i \in \mathcal{T} \cup \mathcal{G}$ and the flip changes the unique failed condition of clause i from false to true.

Indeed, clause i is false at x, since $P_{\ell,q}(x) = 0$, and true at x^e. By (SC) at most one of its conditions changed, so exactly one was false at x, and that one was repaired.

Step 4: $|\mathcal{T}| \leq t - 1$. For $j \in \mathcal{T}$ let $m_j \in [r]$ be the position of its unique failed target, so $P_{\ell-1,Q}(x_{j,c}) = 0$ exactly when $c = m_j$. Let $i, j \in \mathcal{T}$ with $i \to j$. Clause i fails no gate, so $P_{\ell-1,Q-q}(x_{j,a(i,j)}) = 0$. Since $Q - q < Q$, (N) gives $P_{\ell-1,Q}(x_{j,a(i,j)}) = 0$. Thus position $a(i,j)$ is a failed target of clause j, which forces $a(i,j) = m_j$. If $|\mathcal{T}| \geq t$, any t-element $I \subseteq \mathcal{T}$ with these m_j would contradict Lemma 2.1.

Step 5: $|\mathcal{G}| \leq 3$, with at most one exceptional vertex. For $j \in \mathcal{G}$ every child of row j has $P_{\ell-1,Q} = 1$. Let $i, j \in \mathcal{G}$ with $i \to j$. The gate child $x_{j,a(i,j)}$ of clause i lies in row j, so $P_{\ell-1,Q} = 1$ there, and (N) gives $P_{\ell-1,Q-q} = 1$ there: this gate of clause i fails. Clause i fails only one gate, so every vertex of $\mathcal{G}$ has at most one out-neighbour in $\mathcal{G}$. With $m = |\mathcal{G}|$, counting the edges within $\mathcal{G}$ by their tails gives

$$\frac{m(m-1)}{2} \leq m, \qquad \text{hence} \qquad m \leq 3.$$

Of any two vertices of $\mathcal{G}$, one has an edge to the other. So at most one vertex of $\mathcal{G}$ has no out-neighbour in $\mathcal{G}$; call it *exceptional* if it exists.

Step 6: $S_{\ell,0} \le tS'_0 + S'_1 + 3J'_1$. By Step 3, every flip with $P_{\ell,q}(x^e) = 1$ repairs the unique failed condition of some $i \in \mathcal{T} \cup \mathcal{G}$. We bound the number of such flips for each i.

- $i \in \mathcal{T}$: the flip changes $P_{\ell-1,Q}$ on child x_{i,m_i} from 0 to 1; at most S'_0 flips by (C1). Total at most $(t-1)S'_0$ by Step 4.
- $i \in \mathcal{G}$ not exceptional, say $i \to j$ with $j \in \mathcal{G}$: by Step 5 the unique failed gate of clause i is at $y = x_{j,a(i,j)}$, where before the flip $P_{\ell-1,Q-q}(y) = P_{\ell-1,Q}(y) = 1$. The repair makes $P_{\ell-1,Q-q}(y) = 0$, hence $P_{\ell-1,Q}(y) = 0$ by (N). The indices $Q-q$ and Q are distinct, so by (C2) there are at most J'_1 such flips. (The flip breaks a target of clause j; this is harmless, since the accepting clause is i, whose targets and other gates are untouched.)
- i exceptional: the flip changes $P_{\ell-1,Q-q}$ on one fixed child from 1 to 0; at most S'_1 flips.

Since $|\mathcal{G}| \le 3$, the total is at most $(t-1)S'_0 + S'_1 + 3J'_1$.

Step 7: $J_{\ell,0} \le tS'_0 + 3J'_1$. Fix $q < q'$ and x with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 0$. Form $\mathcal{T}$ and $\mathcal{G}$ as in Step 3 for the *smaller* index q, and let g, g' be as in Step 2. Let e be a flip after which both parent predicates are 1. Choose a clause i holding at index q' at x^e; the choice may depend on e. By (N) it holds at index q at x^e, so by Step 3 $i \in \mathcal{T} \cup \mathcal{G}$ and e repairs its unique failed condition at index q.

- $i \in \mathcal{T}$: at most S'_0 flips per clause as in Step 6, in total at most $(t-1)S'_0$.
- $i \in \mathcal{G}$, exceptional or not: the unique failed gate at index q is at a child y with $P_{\ell-1,g}(y) = 1$ before the flip, so $P_{\ell-1,g'}(y) = 1$ before the flip by (N). After the flip clause i holds at index q', and its gate for the same edge tests the same child, so $P_{\ell-1,g'} = 0$ on y, hence $P_{\ell-1,g} = 0$ by (N). Both distinct predicates change from 1 to 0: at most J'_1 flips by (C2), and at most $3J'_1$ in total by Step 5. No target value at y is needed here; the second parent index supplies the second child predicate.

Every counted flip falls under one of these cases, so $J_{\ell,0} \le tS'_0 + 3J'_1$.

All bounds hold for each admissible x and each index or pair of indices, so they hold for the maxima. If $H = 1$ there are no pairs $q < q'$, and the two joint inequalities hold because their left sides are zero. □

4 Quantitative bounds and separation

Lemma 4.1. *Suppose that $d \ge 1$ and $M \ge 9^{d+1}$. For every $0 \le \ell \le d$ and $b \in \{0,1\}$, the predicates of the construction satisfy*

$$S_{\ell,b} \le 2LM^{\ell+b}.$$

It is used at $\ell = d$ in the proof of Theorem 1.1.

Proof. Plan. Normalise $S_{\ell,b}$ and $J_{\ell,b}$ by $M^{\ell+b}$. This turns (3) into a two-term linear recurrence with a small parameter ϵ, which is solved by induction.

Step 1: normalisation. Define

$$u_\ell = \max_{b\in\{0,1\}} \frac{S_{\ell,b}}{M^{\ell+b}}, \qquad v_\ell = \max_{b\in\{0,1\}} \frac{J_{\ell,b}}{M^{\ell+b}}, \qquad \epsilon = \frac{\max(r,t)}{M}.$$

By (2), $u_0 \le \max(L, LM/M) = L$ and $v_0 = 0$.

Step 2: normalised recurrences. Let $1 \le \ell \le d$ and write $u = u_{\ell-1}$, $v = v_{\ell-1}$. By definition $S'_0 \le uM^{\ell-1}$, $S'_1 \le uM^\ell$, $J'_0 \le vM^{\ell-1}$ and $J'_1 \le vM^\ell$. Substituting into (3):

$$S_{\ell,1} \le M^2 uM^{\ell-1} + ruM^\ell = \Big(1 + \frac{r}{M}\Big)u\,M^{\ell+1}, \quad S_{\ell,0} \le tuM^{\ell-1} + uM^\ell + 3vM^\ell = \Big(\Big(1+\frac{t}{M}\Big)u + 3v\Big)M^\ell,$$

$$J_{\ell,1} \le M^2 vM^{\ell-1} + ruM^\ell = \Big(\frac{r}{M}u + v\Big)M^{\ell+1}, \quad J_{\ell,0} \le tuM^{\ell-1} + 3vM^\ell = \Big(\frac{t}{M}u + 3v\Big)M^\ell.$$

Since $r/M, t/M \le \epsilon$ and $v \le 3v$,

$$u_\ell \le (1+\epsilon)u_{\ell-1} + 3v_{\ell-1}, \qquad v_\ell \le \epsilon u_{\ell-1} + 3v_{\ell-1}. \tag{4}$$

Step 3: size of ϵ. Since $M \ge 9^{d+1} \ge 81$, we have $\sqrt{M} \ge 9$, so $r = \lceil\sqrt{M}\rceil \le \sqrt{M}+1 \le 2\sqrt{M}$ and $t = 16 \le 2\sqrt{M}$. Hence

$$\epsilon \le \frac{2}{\sqrt{M}} \le \frac{2}{3^{d+1}}. \tag{5}$$

Step 4: comparison sequences. Let $U_0 = L$, $V_0 = 0$ and

$$U_\ell = (1+\epsilon)U_{\ell-1} + 3V_{\ell-1}, \qquad V_\ell = \epsilon U_{\ell-1} + 3V_{\ell-1}.$$

The coefficients are nonnegative, so (4) and induction give $u_\ell \le U_\ell$ and $v_\ell \le V_\ell$. We claim that for $0 \le \ell \le d$

$$U_\ell \le 2L, \qquad V_\ell \le \epsilon L(3^\ell - 1). \tag{6}$$

This holds at $\ell = 0$. Assume it for all indices below some $1 \le \ell \le d$. First,

$$V_\ell \le 2\epsilon L + 3\epsilon L(3^{\ell-1} - 1) = \epsilon L(3^\ell - 1).$$

Next, $U_j - U_{j-1} = \epsilon U_{j-1} + 3V_{j-1}$, and summing these increments gives

$$\begin{aligned} U_\ell &= L + \sum_{j=0}^{\ell-1}(\epsilon U_j + 3V_j) \le L + 2\epsilon L\ell + 3\epsilon L\sum_{j=0}^{\ell-1}(3^j - 1) \\ &= L + 2\epsilon L\ell + 3\epsilon L\Big(\frac{3^\ell - 1}{2} - \ell\Big) = L + \epsilon L\Big[\frac{3}{2}(3^\ell - 1) - \ell\Big]. \end{aligned}$$

By (5) and $\ell \le d$,

$$\epsilon\Big[\frac{3}{2}(3^\ell - 1) - \ell\Big] \le \frac{2}{3^{d+1}} \cdot \frac{3^{\ell+1} - 3 - 2\ell}{2} = \frac{3^{\ell+1} - 3 - 2\ell}{3^{d+1}} < 1,$$

so $U_\ell < 2L$. This completes the induction. Finally $S_{\ell,b} \le u_\ell M^{\ell+b} \le U_\ell M^{\ell+b} \le 2LM^{\ell+b}$. □

Proof of Theorem 1.1. Fix an integer $d \ge 1$ and an integer $M \ge 9^{d+1}$ (so $M \ge 3$), and labels as in Lemma 2.1. Following the OR-balancing principle of Ambainis and Sun [AS11, Section 4, Lemma 1], let f be the OR of M copies of $P_{d,1}$ on disjoint sets of variables. It is a total Boolean function on

$$n = M \cdot N_d = LM^2(kr)^d \tag{7}$$

bits. Recall $H_d = 1$, so $S_{d,b}$ refers to $P_{d,1}$ alone.

Step 1: $f(0) = 0$ and f is nonconstant. By Lemma 2.3, each copy is 0 at its all-zero input, and flipping one of its blocks makes that copy, hence f, equal to 1.

Step 2: sensitivity. Let $f(x) = 0$. Then every copy is 0, and a flip changes f only if it turns its own copy to 1; there are at most $S_{d,0}$ such flips in each of the M copies. Let $f(x) = 1$. If at least two copies accept, no single flip makes f zero. If exactly one accepts, a sensitive flip must lie in that copy and turn it to 0, giving at most $S_{d,1}$ flips. By Lemma 4.1,

$$s(f) \le \max\{MS_{d,0}, S_{d,1}\} \le 2LM^{d+1}. \tag{8}$$

Step 3: block sensitivity. By Lemma 2.3, each copy has Mk^d pairwise disjoint blocks, each of which turns that copy, hence f, to 1 when flipped at zero. Blocks in different copies are disjoint, so

$$\mathrm{bs}(f, 0) \ge M^2k^d. \tag{9}$$

Step 4: the ratio. Since f is nonconstant, $s(f) > 0$. Using $k/M^2 > 2$ and $L = d + 2$,

$$\frac{\mathrm{bs}(f)}{s(f)^2} \geq \frac{M^2 k^d}{4L^2 M^{2d+2}} = \frac{(k/M^2)^d}{4L^2} \geq \frac{2^d}{4(d+2)^2}. \tag{10}$$

The right side tends to infinity with d. Given $C > 0$, choose d with $2^d/(4(d+2)^2) > C$, then M, labels and f as above; then $\mathrm{bs}(f) > C\, s(f)^2$. □

Sanity check. The bound $2^d/(4(d+2)^2)$ is $256/400 < 1$ at $d = 8$ and $512/484 > 1$ at $d = 9$; this is why the corollary below uses $d = 9$.

4.1 A fixed exponent by composition

Lemma 4.2. *Let $f : \{0,1\}^{n_1} \to \{0,1\}$ and $g : \{0,1\}^{n_2} \to \{0,1\}$ be total Boolean functions. Their disjoint composition is the function on $n_1 n_2$ bits defined by*

$$(f \circ g)(x_1, \ldots, x_{n_1}) = f\big(g(x_1), \ldots, g(x_{n_1})\big), \qquad x_i \in \{0,1\}^{n_2}.$$

Then

$$s(f \circ g) \leq s(f) s(g).$$

If $g(0) = 0$, then $(f \circ g)(0) = f(0)$ and

$$\mathrm{bs}(f \circ g, 0) \geq \mathrm{bs}(f, 0)\, \mathrm{bs}(g, 0).$$

More explicitly, any disjoint families of p_1 and p_2 blocks sensitive at zero for f and g, respectively, yield a disjoint family of $p_1 p_2$ blocks sensitive at zero for $f \circ g$.

This is the induction step in Corollary 4.3.

Proof. Sensitivity. Fix $x = (x_1, \ldots, x_{n_1})$ and put $y = (g(x_1), \ldots, g(x_{n_1}))$. Flipping a coordinate e of child x_i replaces y by y^i if $g(x_i^e) \neq g(x_i)$ and leaves y unchanged otherwise. So e is sensitive for $f \circ g$ at x exactly when e is sensitive for g at x_i and i is sensitive for f at y. Therefore

$$s(f \circ g, x) = \sum_{i:\, f(y^i) \neq f(y)} s(g, x_i) \leq s(f, y) s(g) \leq s(f) s(g).$$

Blocks. Let $g(0) = 0$. At the all-zero input $y = 0$, so $(f \circ g)(0) = f(0)$. Let $\Gamma_1, \ldots, \Gamma_{p_1} \subseteq [n_1]$ and $\Delta_1, \ldots, \Delta_{p_2} \subseteq [n_2]$ be pairwise disjoint sensitive blocks at zero for f and g. Identify the coordinates of $f \circ g$ with $[n_1] \times [n_2]$ and consider the $p_1 p_2$ sets $\Gamma_i \times \Delta_j$. They are nonempty and pairwise disjoint, because the Γ_i are disjoint and the Δ_j are disjoint. Flipping $\Gamma_i \times \Delta_j$ at zero turns each child indexed by Γ_i into 0^{Δ_j}, where $g = 1$, and leaves the other children all-zero, where $g = 0$. The outer input becomes 0^{Γ_i}, and $f(0^{\Gamma_i}) \neq f(0)$. So each $\Gamma_i \times \Delta_j$ is sensitive at zero. Taking maximum families gives the inequality; it is trivial if either maximum is zero. □

Corollary 4.3. *There exist a real number $\alpha > 2$ and nonconstant total Boolean functions F_m, for integers $m \geq 1$, such that*

$$\mathrm{bs}(F_m, 0) \geq s(F_m)^\alpha \qquad \text{for every } m, \qquad \mathrm{bs}(F_m, 0) \longrightarrow \infty.$$

Proof. Seed. Fix $d = 9$, so $L = 11$, an integer $M \geq 9^{10}$ and admissible labels, and let f be the function from the proof of Theorem 1.1. Put

$$A = 22M^{10}, \qquad B = M^2k^9.$$

Then $f(0) = 0$, $s(f) \leq A$ by (8), and $\mathrm{bs}(f, 0) \geq B$ by (9). Moreover

$$\frac{B}{A^2} = \frac{M^2k^9}{484M^{20}} = \frac{(k/M^2)^9}{484} \geq \frac{2^9}{484} = \frac{128}{121} > 1.$$

So $A > 1$ and $B > A^2$, and $\alpha = \log B / \log A$ is a fixed number greater than 2 with $A^\alpha = B$.

Iteration. Set $F_1 = f$ and $F_{m+1} = f \circ F_m$. By induction on m, using Lemma 4.2 with $g = F_m$: F_m is a total function on n^m bits, with n as in (7); $F_m(0) = 0$ (which allows the next application of the lemma); and

$$s(F_m) \leq A^m, \qquad \mathrm{bs}(F_m, 0) \geq B^m.$$

The second bound shows that F_m is nonconstant and that $\mathrm{bs}(F_m, 0) \to \infty$. Since $\alpha > 0$,

$$s(F_m)^\alpha \leq A^{m\alpha} = B^m \leq \mathrm{bs}(F_m, 0).$$

The same estimates give the explicit ratio bound

$$\frac{\mathrm{bs}(F_m)}{s(F_m)^2} \geq \left(\frac{B}{A^2}\right)^m \geq \left(\frac{128}{121}\right)^m. \qquad \square$$

5 Verification checklist

No step is marked unverified; each was rechecked while rewriting. A sceptical reader should check these first.

1. **Direction of nesting (N)**, Section 2. Smaller index means weaker predicate. Every use applies it to two predicates *on the same child*: a passing gate forces a failed target (Step 4), and a true target forces a failed gate (Step 5).
2. **(SC)**: the $r + M^2$ conditions of one clause test distinct children. This is what restricts attention to clauses with exactly one failed condition (Step 3).
3. **Lemma 2.1**: 120 independent equations against $k^{16}r^{16}$ patterns, giving $3^{16}M^{-20} < 1$. Its use in Step 4 needs $a(i, j) = m_j$ for *every* edge inside $\mathcal{T}$.
4. **Step 5**: outdegree at most one inside $\mathcal{G}$ gives $|\mathcal{G}| \leq 3$ and at most one exceptional vertex.
5. **Steps 6 and 7**: a non-exceptional gate repair changes $P_{\ell-1,Q-q}$ and $P_{\ell-1,Q}$ together; in the joint case the pair is $P_{\ell-1,g}$, $P_{\ell-1,g'}$ and no vertex is exceptional. Check that the index pairs are distinct and lie in $[Q]$.
6. **Lemma 2.2(iv)**: $J_{0,b} = 0$. Without it $v_0 \neq 0$ and the induction in Lemma 4.1 fails.
7. **Lemma 4.1**: the powers of M in Step 2, the bound $\epsilon \leq 2/3^{d+1}$ in Step 3, and the final inequality of Step 4.
8. **Proof of Theorem 1.1**: the case analysis $s(f) \leq \max\{MS_{d,0}, S_{d,1}\}$ and the arithmetic in (10).
9. **Lemma 4.2**: the block bound uses $g(0) = 0$, and the induction in Corollary 4.3 must maintain $F_m(0) = 0$.

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